

VBF-T9R4-BOM-001 Bill of Materials

t9_r4 - grid-storage cell cathode on the Chen2020 NMC811/graphite profile

Result: HONEST NEGATIVE (no candidate satisfies the full contract; nothing reported as passing)

Fixed cell stack (Chen2020 baseline, unchanged by design - the cathode composition is the only free variable):

Item	Specification	Source
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Positive current collector	Al, 16 um	Chen2020
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Negative current collector	Cu, 12 um	Chen2020
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Positive electrode coating	75.6 um, eps_am 0.665, rho 3262 kg/m3 (NMC811 baseline)	Chen2020
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Negative electrode	85.2 um, eps_am 0.75, graphite rho 1657 kg/m3	Chen2020
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Separator	12 um, porosity 0.47	Chen2020
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Electrolyte	EC/EMC + LiPF6 (anodic limit 4.8 V) - FIXED	contract
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Cathode active material	NOT RESOLVED - no candidate passed; screening space: 42 layered LiMO2 compositions	negative result
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Electrode area	0.1027 m2 (0.065 x 1.58 m); nominal 5.0 Ah	Chen2020
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Screened cathode-active candidates (3 run-comp batches, all layered LiMO2 prototypes):

round 1: LiCuO2, LiFeO2, CuNi55, CuMn55, CuFe55, CuCo55, NiFe55, CoFe55, FeMn55, LiVO2, CuAl91, CuZr91, CuMn73, CuNi82; round 2: LiScO2, LiAlO2, LiGaO2, LiZnO2, FeNi73, FeNi37, CrFe55, CrNi55, CrCo55, FeCo91, NiFe91, FeNi91, ZnNi55, ScFe55; round 3: LiBO2, AlB55, AlB73, AlB91, BaI73, BaI91, AlB82, AlB64, AlB95, GaAl55, AlSc55, BGa55, BSc55, GaAl91.

All rejected (window, compatibility, or both).